Response Style Variation Across Model Families: Evidence from Large-Scale Psychometric Probing of LLMs

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Abstract

A growing body of work applies personality inventories to LLMs, yet it remains unclear whether the resulting scores capture meaningful individual differences or systematic response tendencies shaped by training and alignment. In this study, we administer 61 Likert-scale items spanning 9 psychometric dimensions to one model from each of 15 model families via unified APIs, collecting 10,980 responses (12 seeds per model), and characterize the resulting response style variation. We find that model families exhibit large, systematic differences in their response distributions (median pairwise Cohen's $d = 1.01$), with Neuroticism showing the largest between-family effect ($\eta_p^2 = 0.75$). The inter-dimension correlation structure reorganizes relative to humans: the Conscientiousness–Neuroticism correlation reverses from $r = -0.25$ to $r = +0.76$ (95% CI $[0.47, 0.92]$), and Extraversion–Openness flips from positive to negative. Convergent validity against published human baselines yields insufficient evidence for construct equivalence (3/5 sign agreement, $p = 0.50$), further indicating that standard dimension labels do not transfer cleanly to LLMs. Together, these findings demonstrate systematic response style variation across LLM families and quantify the degree to which psychometric responses are shaped by training and alignment rather than by human-like individual differences.

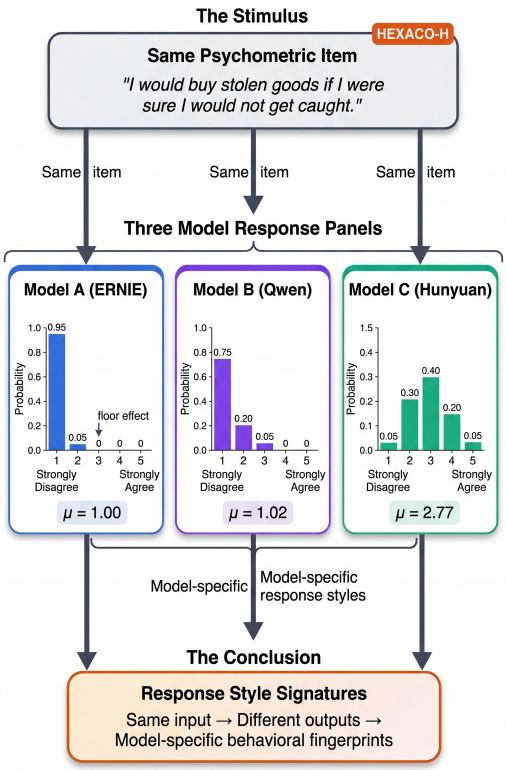

Figure 1: **Core concept.** Identical psychometric items elicit systematically different Likert-scale responses from different models. These model-specific response style signatures raise the question: do psychometric scores in LLMs reflect personality traits, or systematic response tendencies shaped by training and alignment?

1 Introduction

A growing number of studies apply personality inventories to LLMs, reporting that different models exhibit different trait profiles (Jiang and Althoff, 2023; Pan and Zeng, 2024; Serapio-García et al., 2025). These results are often interpreted as evidence that LLMs possess personality-like traits shaped by training and alignment. However, if LLMs respond to personality items using surface-level heuristics—such as mimicking socially desirable answers or following alignment-induced response tendencies—then the resulting scores may capture systematic *response styles* rather than personality traits, and the inter-dimension correlations may differ fundamentally from the human factor structure. Following Binz and Schulz (2023), we use the term *response style* to describe observed response patterns without claiming they reflect personality traits or inner states.

We characterize this response style variation with a large-scale cross-model comparison (Fig-

ure 1). Using unified API platforms, we administer 61 Likert-scale items spanning 9 dimensions—the Big Five Inventory (John and Srivastava, 1999), HEXACO Honesty-Humility (Lee and Ashton, 2004), Schwartz Values (Schwartz, 1992), cognitive style, and cultural dimensions—to one model from each of 15 model families, repeating each query across 12 random seeds (10,980 total API calls). We also track 16 additional models from three families (Qwen, DeepSeek, GLM) across versions, yielding 31 models total.

We present three findings that characterize systematic response style variation across LLM families:

1. **Model families exhibit distinct response style signatures.** Families differ substantially in their response distributions (median pairwise Cohen's $d = 1.01$), indicating that different training approaches produce systematic behavioral fingerprints.
2. **Inter-dimension correlations reorganize relative to humans.** The Conscientiousness–Neuroticism correlation reverses from $r = -0.25$ in humans to $r = +0.76$ in LLMs (95% CI $[0.47, 0.92]$), and Extraversion–Openness flips from positive to negative, patterns observed across most tested subgroups.
3. **Standard dimension labels do not transfer cleanly.** Comparison against published human baselines yields only 3/5 sign agreement ($p = 0.50$); this inconclusive convergent validity result itself constitutes evidence for trait structure reorganization rather than random noise.

We test convergent validity against human baselines (Appendix E), correct for acquiescence bias (§4.6), and report 95% confidence intervals for key statistics.

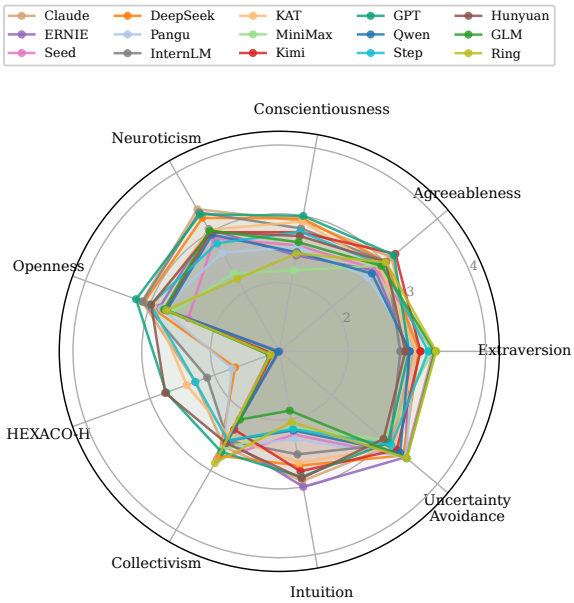

Figure 2: Response style profiles of all 15 model families across 9 dimensions. Each line is one model family. The distinct shapes suggest that different training approaches leave different behavioral fingerprints on models.

2 Related Work

LLM personality and response behavior. Jiang and Althoff (2023) found temperature-dependent personality variation in ChatGPT using the BFI-10. Pan and Zeng (2024) compared LLMs from different cultural backgrounds on the BFI-44, reporting cultural effects. Myska and Macková (2023) showed that GPT-3.5 and GPT-4 produce stable profiles across sessions, and Schaeckermann et al. (2024) examined 9 LLMs on the Big Five, finding moderate-to-large inter-model differences. Serapio-García et al. (2025) proposed a psychometric framework for evaluating and shaping personality traits across 18 LLMs, reporting that larger models exhibit broader trait ranges. Hilliard et al. (2024) confirmed that model scale affects elicited personality profiles, with larger models showing more extreme trait scores. Bodroža et al. (2024) tested 7 LLMs on multiple personality instruments, finding limited temporal stability but consistently elevated prosociality. Santos and Costa (2024) surveyed personality assessment with LLMs, highlighting both promises and pitfalls of treating model outputs as psychometric data. Sorokovikova et al. (2024) replicated Big Five profiles across three models and found systematic deviations from human norms, while Tosato et al. (2025) documented persistent instability across scales and conversation histories.

While Serapio-García et al. (2025) evaluated the largest set of models to date (18), their study used heterogeneous APIs and did not test for structural differences in inter-dimension correlations. Our work contributes unified API control across 15 model families and explicit validity testing against human baselines.

Behavioral evaluation beyond capabilities. Standard capability benchmarks (Hendrycks et al., 2021; Liang et al., 2022) measure task performance—reasoning, coding, instruction-following—and instruction-following evaluators such as AlpacaEval (Dubois et al., 2024) assess response quality, but none address how models respond to self-referential statements about preferences and moral tendencies. Laverghetta et al. (2022) showed that language models can predict human psychometric properties from text, raising the question of whether LLM self-reports reflect similar constructs. We contribute a behavioral evaluation dimension: how models respond to standardized self-referential questions.

Response biases and measurement validity. Applying human-designed psychometric instruments to LLMs raises validity concerns (Schaeckermann et al., 2024; Serpell and Ney, 2024). Sühr et al. (2024) question whether standard personality tests are valid for LLMs at all, given the lack of genuine self-report. Salecha and Kosinski (2024) demonstrated that LLMs exhibit human-like social desirability biases in Big Five survey responses, inflating socially desirable traits. Domínguez-Olmedo et al. (2024) showed that LLM survey responses lack the natural variation seen in human populations, producing compressed response distributions. Tjuatja et al. (2023) found that LLMs fail to replicate known human response biases in survey design. Braun (2025) documented acquiescence bias in LLMs, systematically agreeing with questionnaire statements regardless of content. Han et al. (2025) demonstrated a dissociation between LLM self-reported personality and behavioral choices, reinforcing the need to distinguish stated preferences from actions. Xie et al. (2025) introduced AIPsychoBench, a benchmark specifically designed to measure psychometric differences between LLMs and humans, finding systematic discrepancies in response patterns. We address these concerns by framing measurement explicitly as "response style" rather than personality, testing convergent validity against human data, and correcting for acquiescence bias—extending methodological transparency beyond prior work.

Factor structure in LLMs. The Five-Factor Model (McCrae and Costa, 1992) provides the theoretical foundation for the BFI dimensions we use. Suh and Moon (2024) recovered the Big Five factor structure from LLMs using singular value decomposition on log-probabilities of personality-relevant adjectives. Their approach analyzes semantic space, whereas we analyze questionnaire responses—two complementary probes. An exploratory PCA on our data hints at a three-factor reorganization (Appendix D), but this is severely underpowered ($n = 15$).

3 Method

3.1 Psychometric Instruments

We use 9 dimensions comprising 61 Likert items (1–5 scale; Table 1). Five come from the Big Five Inventory (John and Srivastava, 1999; McCrae and Costa, 1992; O'Connor, 2007): Extraversion (8 items), Agreeableness (9), Conscientiousness (9), Neuroticism (8), and Openness (10). We add HEXACO Honesty-Humility (5 items; Lee and Ashton, 2004), Collectivism (4 items; Schwartz Values Survey Schwartz, 1992), Intuition (4 items; cognitive style), and Uncertainty Avoidance (4 items; cultural dimensions). Reverse-scored items are transformed via score $= 6 -$ original; dimension scores are the mean of constituent items.

Table 1: Psychometric instruments: 9 dimensions, 61 Likert items (1–5 scale).

Dimension	Source	Items	Reverse Scored
Extraversion	BFI-44	8	4 items
Agreeableness	BFI-44	9	4 items
Conscientiousness	BFI-44	9	4 items
Neuroticism	BFI-44	8	2 items
Openness	BFI-44	10	3 items
HEXACO-H	HEXACO	5	0
Collectivism	Schwartz	4	0
Intuition	Cognitive Style	4	2 items
Uncertainty Avoidance	Cultural Dimensions	4	2 items
Total		**61**	

3.2 What We Measure: Response Style

We frame our measurements as **response style** rather than personality (Paulhus, 1991; Rammstedt and John, 2000). In psychometrics, response style refers to systematic tendencies in how respondents answer questionnaire items, independent of item content: acquiescence (yea-saying), midpoint

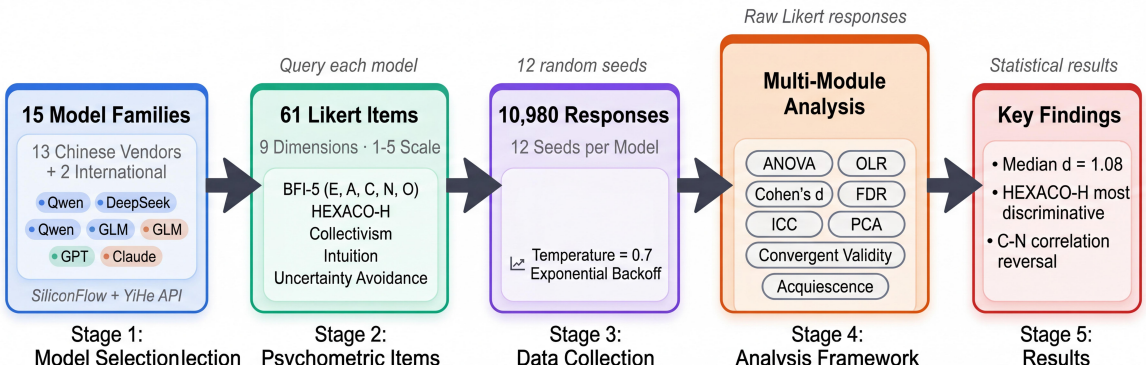

Figure 3: **Experimental pipeline.** We select 15 model families via unified API platforms, query each model on 61 Likert items (9 dimensions, 12 seeds), collect 10,980 responses, and apply a multi-module analysis framework (ANOVA, Cohen's d, convergent validity, reliability, acquiescence correction) to characterize model-level response style variation.

responding, extreme responding, and socially desirable responding (Baumgartner and Steenkamp, 2007). These styles are properties of the *respondent*, not of the construct being measured—a respondent who agrees with every statement is not necessarily more extraverted or agreeable.

We adopt this framing because LLMs lack the subjective experience that personality constructs presume. When an LLM reports high Conscientiousness, this reflects a systematic tendency in its training data and alignment to produce conscientiousness-consistent responses, not a genuine psychological trait. Our research question is therefore descriptive: *do models from different families exhibit systematically different response style signatures when presented with identical psychometric items?* We report dimension labels ("Extraversion," "Neuroticism") for interpretability but explicitly test (and fail to establish) that these labels capture constructs equivalent to human personality (Appendix E).

3.3 Model Selection

We select 15 model families accessible through unified API platforms, contributing one representative model each (Table 2). Priorities: flagship models where available; Dense and MoE architectures both represented (10 Dense, 5 MoE).

3.4 Experimental Protocol

Each model is administered all 61 items per seed at temperature $= 0.7$, with 12 seeds (0, 1, 2, 4, 5, 6, 7, 8, 9, 42, 123, 456) capturing response distribution variation, totaling 10,980 API calls. We set temperature to 0.7 to balance response diversity with reproducibility: temperature 0 yields deterministic outputs that eliminate within-model variance, while higher temperatures (≥ 1.0) may produce inconsistent self-referential responses. Prior work on LLM personality (Jiang and Althoff, 2023) has shown that temperature affects reported trait scores, making a moderate temperature necessary to capture the full range of each model's response tendencies. Items are framed as self-referential statements (*e.g.* "I see myself as someone who is the life of the party") with a 1–5 agreement scale. Rate limiting uses exponential backoff on 429 responses with up to 5 concurrent threads.

Table 2: Models from 15 model families, one representative model per family. Total parameters in parentheses where publicly disclosed.

#	Model Family	Model ID	Architecture
1	Qwen	Qwen3.5-397B-A17B	MoE (397B)
2	DeepSeek	DeepSeek-V3.2	MoE (671B)
3	GLM	GLM-5	MoE (744B)
4	Kimi	Kimi-K2.5	MoE (1.1T)
5	ERNIE	ERNIE-4.5-300B-A47B	MoE (300B)
6	Hunyuan	Hunyuan-A13B-Instruct	Dense (13B)
7	Seed	Seed-OSS-36B-Instruct	Dense (36B)
8	InternLM	internlm2_5-7b-chat	Dense (7B)
9	Ring	Ring-flash-2.0	MoE (100B)
10	Step	Step-3.5-Flash	MoE (197B)
11	Pangu	pangu-pro-moe	MoE (72B)
12	KAT	KAT-Dev	Dense (32B)
13	MiniMax	MiniMax-M2.5	MoE (230B)
14	GPT	GPT-5	Undisclosed
15	Claude	Claude Opus 4.5	Undisclosed

3.5 Analysis Framework

Figure 3 illustrates the full experimental pipeline. Our analysis combines several complementary approaches. For each dimension, we run a one-way ANOVA (Model Family, 15 levels) with FDR-corrected p-values (Benjamini and Hochberg,

1995) and partial η_p^2, supplemented by ordered logistic regression to account for the ordinal nature of Likert responses. The effect size is quantified as partial η_p^2:

$$\eta_p^2 = \frac{SS_{\text{between}}}{SS_{\text{between}} + SS_{\text{within}}} \quad (1)$$

where SS_{between} and SS_{within} are the between-group and within-group sums of squares, respectively. Pairwise Cohen's d (Welch's t-test) is computed for all 105 model pairs across 8 primary dimensions (840 tests, FDR-corrected):

$$d_{ij} = \frac{\bar{x}_i - \bar{x}_j}{s_{\text{pooled}}}, \quad s_{\text{pooled}} = \sqrt{\frac{(n_i-1)s_i^2 + (n_j-1)s_j^2}{n_i + n_j - 2}} \quad (2)$$

where $\bar{x}_i$ and $\bar{x}_j$ are the mean dimension scores for model families i and j, s_i and s_j are their standard deviations, n_i and n_j are the number of observations (12 seeds each), and s_{pooled} is the pooled standard deviation. Power analysis confirms adequate power ($> 80\%$) for $d \geq 0.8$ with 12 observations per model. To correct for acquiescence bias, we compute partial correlations controlling for the first principal component (PC1):

$$r_{XY \cdot Z} = \frac{r_{XY} - r_{XZ} r_{YZ}}{\sqrt{(1 - r_{XZ}^2)(1 - r_{YZ}^2)}} \quad (3)$$

where r_{XY} is the Pearson correlation between dimensions X and Y, r_{XZ} and r_{YZ} are their respective correlations with PC1 (Z), and $r_{XY \cdot Z}$ is the partial correlation between X and Y after removing the variance explained by PC1. We assess reliability via ICC(1,1) and coefficient of variation, classify dimensions with inter-model $\sigma < 0.15$ as alignment artifacts, and compare LLM inter-dimension correlations against published human baselines (John and Srivastava, 1999) for 5 key dimension pairs. For all correlation and ANOVA analyses, we first aggregate the 12 seed-level responses per model into dimension means, then compute statistics at the model-family level ($n = 15$). The human baseline correlations are drawn from the large-scale validation of the BFI-44 reported in the BFI-2 manual (O'Connor, 2007) and the trait taxonomy chapter (John and Srivastava, 1999); exact standard deviations for these inter-dimension correlations are not always available in the original sources.

4 Experiments

4.1 Overview

Table 3 reports model family mean scores across 9 dimensions. On the Big Five, scores cluster around the scale midpoint (2.7–3.3) with the exception of Neuroticism, where Ring (2.22) and MiniMax (2.31) stand apart from GPT (3.30) and Claude (3.39). Intuition shows the widest spread among non-H dimensions: GLM scores lowest (1.88) while Claude leads (2.92), a gap of over one full point. Collectivism and Uncertainty Avoidance are the most compressed dimensions (ranges of 0.73 and 0.44, respectively), suggesting near-uniform responses across model families. In the raw data, Dense models (GPT, Claude, Kimi, Hunyuan, Seed, InternLM, Ring, Step, KAT, MiniMax) tend to score higher on HEXACO-H and Agreeableness than MoE models, while MoE models show higher Uncertainty Avoidance—a pattern we revisit in Appendix F with appropriate caution about confounding between architecture, provider, and scale. Figure 2 overlays all 15 profiles, revealing that models from the same provider tend to share similar profile shapes. Family-level clustering is further illustrated by within-family trajectories across model versions (Appendix J).

4.2 Cross-Model Discrimination

All 8 primary dimensions show significant model effects ($p < 0.001$, FDR-corrected; Table 5), with the strongest discrimination on Neuroticism ($F = 34.60$, $\eta_p^2 = 0.75$) and Intuition ($F = 20.26$, $\eta_p^2 = 0.63$); non-parametric Kruskal-Wallis tests confirm these results. Figure 4 shows the inter-model σ by dimension: among the 8 primary dimensions, Intuition (0.37) and Neuroticism (0.34) show the most variability, while Agreeableness ($\sigma = 0.15$) and UA ($\sigma = 0.16$) fall below our alignment-artifact threshold of 0.15. HEXACO-H has the highest overall variability (0.67) but is excluded from primary analysis ($\alpha = 0.27$); detailed results are in Appendix F.

Architecture type (Dense vs. MoE) shows significant differences on several dimensions (Table 4), with 10 Dense and 5 MoE families. The strongest signals are on Agreeableness ($d = 0.78$, $p < 0.001$) and UA ($d = -0.83$, $p < 0.001$); HEXACO-H shows an even larger apparent effect ($d = 0.91$) but is excluded from primary analysis due to low reliability. **However, architecture is fully confounded with provider** (all MoE models

Table 3: Model family mean scores (SD) across 9 dimensions. Range: 1–5. Primary analysis uses 8 dimensions ($\alpha > 0.70$); HEXACO-H ($\alpha = 0.27$) shown for completeness only.

Model Family	Extraversion	Agreeableness	Conscientiousness	Neuroticism	Openness	HEXACO-H	Collectivism	Intuition	UA
ERNIE	2.89 (0.14)	2.82 (0.07)	2.42 (0.10)	3.05 (0.14)	2.86 (0.14)	1.00 (0.00)	2.50 (0.21)	3.00 (0.15)	3.40 (0.20)
Seed	3.24 (0.26)	2.85 (0.27)	2.56 (0.33)	2.90 (0.25)	2.41 (0.36)	1.15 (0.15)	2.81 (0.26)	2.23 (0.39)	3.40 (0.25)
DeepSeek	3.03 (0.29)	3.03 (0.23)	2.96 (0.14)	3.24 (0.24)	3.09 (0.17)	1.68 (0.34)	2.75 (0.24)	2.69 (0.19)	3.35 (0.23)
Pangu	2.98 (0.24)	2.68 (0.21)	2.55 (0.24)	2.66 (0.29)	2.82 (0.35)	1.73 (0.37)	2.62 (0.25)	2.29 (0.28)	3.40 (0.23)
InternLM	2.76 (0.24)	3.03 (0.14)	2.81 (0.17)	3.33 (0.19)	3.12 (0.24)	2.12 (0.38)	2.50 (0.18)	2.52 (0.23)	3.10 (0.25)
KAT	3.00 (0.00)	2.91 (0.21)	2.92 (0.14)	3.04 (0.11)	2.95 (0.14)	2.43 (0.27)	2.44 (0.19)	2.65 (0.20)	3.17 (0.16)
MiniMax	3.23 (0.34)	2.94 (0.25)	2.19 (0.21)	2.31 (0.22)	2.67 (0.15)	1.08 (0.23)	2.65 (0.27)	2.12 (0.13)	3.00 (0.34)
Kimi	3.05 (0.18)	3.20 (0.12)	2.76 (0.15)	3.00 (0.11)	2.77 (0.17)	1.10 (0.13)	2.31 (0.11)	2.77 (0.31)	3.23 (0.33)
Qwen	2.88 (0.17)	2.74 (0.26)	2.45 (0.11)	2.98 (0.17)	2.71 (0.17)	1.00 (0.00)	2.50 (0.00)	2.10 (0.17)	3.29 (0.14)
Step	3.17 (0.16)	2.95 (0.16)	2.77 (0.26)	2.81 (0.26)	2.99 (0.18)	2.30 (0.64)	2.50 (0.18)	2.15 (0.61)	3.12 (0.23)
Hunyuan	2.83 (0.14)	3.02 (0.09)	2.70 (0.14)	3.00 (0.00)	2.98 (0.14)	2.77 (0.27)	2.54 (0.14)	2.88 (0.13)	2.98 (0.07)
GLM	3.26 (0.26)	2.94 (0.13)	2.61 (0.24)	3.03 (0.19)	2.78 (0.19)	1.15 (0.12)	2.15 (0.13)	1.88 (0.17)	3.42 (0.16)
Ring	3.28 (0.40)	3.02 (0.36)	2.44 (0.40)	2.22 (0.29)	2.75 (0.34)	1.13 (0.26)	2.88 (0.36)	2.04 (0.46)	3.42 (0.33)
GPT	2.89 (0.17)	3.18 (0.29)	3.00 (0.17)	3.30 (0.16)	3.22 (0.37)	2.75 (0.44)	2.69 (0.30)	2.85 (0.27)	3.06 (0.26)
Claude	2.96 (0.18)	3.13 (0.18)	2.93 (0.21)	3.39 (0.19)	3.12 (0.17)	2.30 (0.32)	2.60 (0.13)	2.92 (0.12)	3.12 (0.13)

are from Chinese providers, while Dense models include GPT and Claude) and partially confounded with model scale, so these patterns must be interpreted as descriptive observations of co-variation rather than evidence for causal architecture effects.

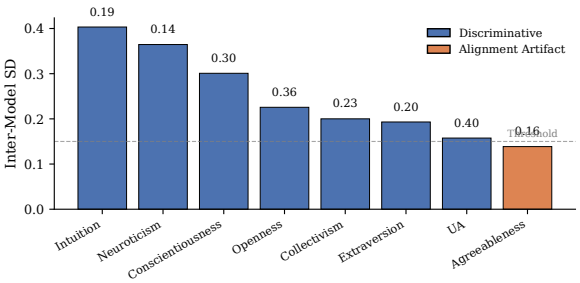

Figure 4: Inter-model SD by dimension. Blue: discriminative ($\sigma \geq 0.15$); orange: alignment artifact.

Table 4: Mean dimension scores by architecture type (8 primary dimensions; HEXACO-H excluded). Dense: $n = 10$ model families (GPT, Claude, Kimi, Hunyuan, Seed, InternLM, Ring, Step, KAT, MiniMax). MoE: $n = 5$ (Qwen, DeepSeek, GLM, ERNIE, Pangu). 95% CI for Cohen's d from bootstrap (5,000 iterations). Effects surviving FDR correction are marked with *; no claim of causal architecture effect is intended due to confounding with provider and scale.

Dimension	Dense	MoE	d (95% CI)	p
Extraversion	3.04	3.01	+0.12 [−0.18, +0.42]	0.42
Agreeableness	3.02	2.84	+0.78 [+0.47, +1.11]	0.000*
Conscientiousness	2.71	2.60	+0.36 [+0.07, +0.70]	0.014
Neuroticism	2.93	2.99	−0.16 [−0.42, +0.12]	0.25
Openness	2.90	2.85	+0.14 [−0.14, +0.43]	0.32
Collectivism	2.59	2.50	+0.32 [+0.01, +0.64]	0.046
Intuition	2.51	2.39	+0.27 [−0.03, +0.60]	0.093
UA	3.16	3.37	−0.83 [−1.15, −0.55]	0.000*

The pairwise effect sizes are large: the median Cohen's d across all 105 model pairs and 8 primary dimensions is 1.01, and every pair exceeds $d \geq 0.8$ on at least one dimension (Figure 5). Table 6 reports the largest pairwise Cohen's d for each dimension. Among the 8 primary dimensions, Neuroticism and Conscientiousness are most separable (max $d = 5.29$ and 4.45), while Agreeableness shows the weakest discrimination (median $d = 0.74$). Notably, GPT and Claude produce the smallest Euclidean distance of any model pair (0.50), suggesting convergence in training objectives across leading model families.

Table 5: ANOVA results: model effect on each dimension ($df = 14,165$). Primary analysis uses 8 dimensions ($\alpha > 0.70$); HEXACO-H excluded ($\alpha = 0.27$, 5 items). All $p < 0.001$ after FDR correction.

Dimension	F	p (raw)	η_p^2	KW H	KW p
Extraversion	6.55	2.0e-10	0.357	65.5	1.3e-08
Agreeableness	5.99	1.8e-09	0.337	62.4	4.4e-08
Conscientiousness	14.35	4.8e-22	0.549	99.1	7.1e-15
Neuroticism	34.60	8.1e-42	0.746	124.7	7.6e-20
Openness	9.95	6.7e-16	0.458	80.2	2.6e-11
Collectivism	9.01	1.9e-14	0.433	78.7	4.9e-11
Intuition	20.26	5.5e-29	0.632	118.5	1.3e-18
UA	5.68	6.3e-09	0.325	62.3	4.6e-08

Table 6: Largest pairwise Cohen's d per dimension (105 pairs each, FDR-corrected). Median d across all pairs shown. HEXACO-H excluded from primary analysis ($\alpha = 0.27$).

Dimension	Model A		Model B	Med d	d
Neuroticism	MiniMax	vs	Claude	1.66	−5.29
Conscientiousness	ERNIE	vs	DeepSeek	1.22	−4.45
Intuition	GLM	vs	Claude	1.53	−7.06
Collectivism	GLM	vs	Qwen	0.80	−3.89
Agreeableness	ERNIE	vs	Kimi	0.74	−3.72
UA	GLM	vs	Hunyuan	0.81	+3.47
Openness	MiniMax	vs	Claude	0.98	−2.85
Extraversion	Step	vs	Hunyuan	0.90	+2.17

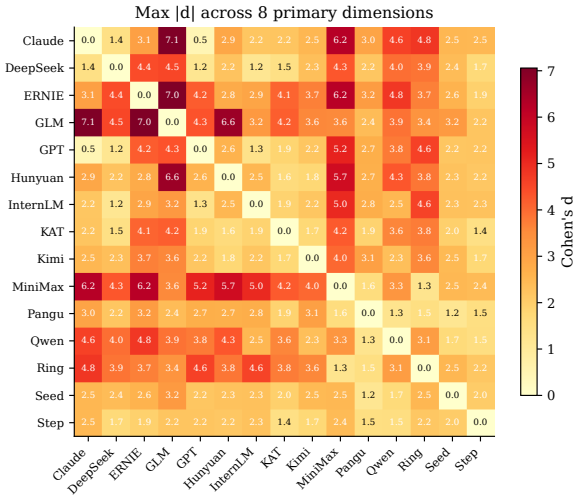

Figure 5: Maximum pairwise Cohen's d per model pair (across 9 dimensions). All pairs exceed $d \geq 0.8$.

4.3 Inter-Dimension Correlations (Exploratory)

The 9×9 model-level correlation matrix (Figure 9) suggests that the human BFI factor structure does not fully recover in LLMs. The most notable divergence is the Conscientiousness–Neuroticism correlation, which is strongly positive in LLMs ($r = +0.76$, 95% CI $[0.47, 0.92]$) but weakly negative in humans ($r = -0.25$), and this reversal holds across multiple subgroups (Appendix E). A convergent validity test against human baselines yields 3/5 sign agreement (binomial $p = 0.50$), providing no evidence that the measured dimensions capture constructs equivalent to human personality (Appendix E).

These structural observations are **exploratory** ($n = 15$ families) and should be interpreted with caution; the wide confidence intervals at this sample size mean that replication with a broader model sample is needed before drawing firm conclusions about factor structure reorganization.

4.4 Aligned vs. Base: Preliminary Evidence

To test whether alignment shapes response style, we compare base (pre-alignment) models against their aligned counterparts from the same provider: Qwen3-8B vs. Qwen3.5-397B, Qwen3-14B vs. Qwen3.5-397B, and GLM-4.7 vs. GLM-5 (Table 7). We acknowledge two limitations of this design: (1) the Qwen comparisons confound alignment with a large scale difference (8B/14B → 397B), and (2) GLM-4.7 may retain partial alignment from its training, making it not a fully "base" model. Despite these caveats, if response style variation reflects inherent model properties rather than alignment, base and aligned models should show similar profiles.

Table 7: Base vs. aligned model comparison (Study 4). Base models lack RLHF/safety alignment; aligned models are their post-alignment counterparts from the same provider. Cohen's d reported as base minus aligned.

Dimension	Qwen3-8B vs 397B	Qwen3-14B vs 397B	GLM-4.7 vs GLM-5
Extraversion	−0.36	+1.30	−1.46
Agreeableness	+2.24	+2.50	−0.83
Conscientiousness	+4.62	+8.67	−1.20
Neuroticism	+3.29	+0.22	−1.09
Openness	+5.51	+2.96	−1.56
HEXACO-H	+11.19	+15.81	+0.85
Collectivism	+2.77	−1.01	+1.20
Intuition	+4.02	+6.79	+2.01
UA	−2.30	−0.51	+0.80

The results provide preliminary evidence that alignment shapes response style variation, though the scale confound in the Qwen comparisons limits causal attribution. Base models produce responses clustered at the Likert scale midpoint (3.0): Qwen3-14B shows exactly zero variance on 7 of 9 dimensions, and Qwen3-8B shows near-zero variance ($\sigma < 0.20$ on all dimensions). In contrast, their aligned counterpart (Qwen3.5-397B) exhibits the differentiated, compressed profiles seen throughout Study 1. Among the 8 primary dimensions, alignment effects are largest on Intuition ($d = 2.01$–6.79) and Conscientiousness ($d = 1.20$–8.67). The most dramatic single-dimension effect is on HEXACO-H ($d > 11$), where base models score 2.15–2.40 and the aligned model scores 1.00; however, given HEXACO-H's low reliability ($\alpha = 0.27$), this finding is suggestive rather than conclusive.

GLM-4.7 vs. GLM-5 shows more moderate effects on the primary dimensions, suggesting that GLM-4.7 retains partial alignment from its training. Taken together, these comparisons suggest that post-training alignment substantially shapes response style, but the scale–alignment confound means this evidence is preliminary and should be replicated with same-scale model pairs.

4.5 Within-Family Evolution

To examine whether response styles change across model versions, we track 16 additional models from three families (Qwen: 6 versions; DeepSeek: 4 versions; GLM: 6 versions; full trajectories in Appendix J). Most dimensions show moderate variation without clear monotonic trends. Notable exceptions: DeepSeek shows a steep decline on HEXACO-H from V2.5 ($H = 3.00$) to V3.2

($H = 1.68$), consistent with intensified safety alignment between versions (though HEXACO-H is excluded from primary analysis); GLM shows the largest within-family shift on Intuition (5.0 to 3.875). These observations are descriptive and motivate future controlled experiments.

4.6 Acquiescence and Reliability

Response style indicators. Following Paulhus (1991), we operationalize three response style indicators directly from item-level responses (Figure 6). **Midpoint responding** (proportion of responses at the scale midpoint, 3.0) varies widely across families: Step (83%) and KAT (87%) show extreme midpoint bias, while MiniMax (20%) and Claude (27%) show low midpoint responding. **Extreme responding** (proportion of responses at 1 or 5) shows the opposite pattern: Ring (53%) and Seed (43%) produce the most extreme responses, while Hunyuan (2%) and Claude (4%) rarely use scale endpoints. **Acquiescence bias** (mean of positively worded items minus mean of reverse-scored items) is consistently negative across 14 of 15 families (mean $= -0.28$), indicating a slight tendency to disagree with positively worded items and agree with negatively worded items; Ring is the sole exception ($+0.24$). These indicators confirm that models differ substantially in how they use the Likert scale, consistent with the response style framework.

Acquiescence factor. The first principal component of the 15-family correlation matrix explains 51.1% of variance (vs. 11.1% expected under independence), indicating a strong general factor. Partial correlations controlling for PC1 reduce median absolute correlation by 45%, but 16 of 36 pairs show sign flips, indicating that the general factor is entangled with the dimension structure rather than representing a simple acquiescence bias.

Reliability. Table 8 summarizes reliability metrics. Within-model CV is uniformly low (0.065–0.108 for the 8 primary dimensions), indicating tight response distributions, while ICC(1,1) ranges from 0.281 to 0.737, with the highest value on Neuroticism (0.737), reflecting large between-family differences.

4.7 Prompt Sensitivity

If LLM psychometric scores reflect stable traits, they should be invariant to surface-level changes in question framing. We administered the same 61 Likert items to all 15 Study 1 models under four prompt conditions (12 seeds each, temperature $= 0.7$): **Default** ("respond as honestly as possible"), **Neutral** (instruction removed, "rate your agreement"), **Persona** (human role-play: "you are completing a personality survey"), and **Direct** (no instruction, only the item and scale). Two models (GLM-5, Ring-flash-2.0) were unavailable for the Direct condition due to API downtime; 13 models have complete data across all four conditions.

Table 8: Reliability across all 15 models. CV = coefficient of variation across seeds. ICC(1,1) = intraclass correlation for absolute agreement, computed at the family level ($n = 15$ families $\times$ 12 seeds). HEXACO-H ($\alpha = 0.27$) shown for completeness but excluded from primary analysis.

Dimension	Mean CV	ICC(1,1)
Extraversion	0.069	0.316
Agreeableness	0.067	0.293
Conscientiousness	0.076	0.527
Neuroticism	0.065	0.737
Openness	0.077	0.427
HEXACO-H	0.146	0.824
Collectivism	0.076	0.400
Intuition	0.108	0.616
UA	0.068	0.281

Score shifts. Within-model standard deviation across the four prompt conditions averaged 0.127 on the 1–5 scale (range 0.083–0.174 across dimensions). Conscientiousness and Neuroticism showed the largest prompt sensitivity (mean SD $= 0.173$ and 0.174), while Agreeableness and Uncertainty Avoidance were most stable (mean SD $= 0.083$ and 0.093). Cohen's d between each variant and the Default condition yielded a median $|d|$ of 0.49–0.98 across dimensions; 75% of model-dimension pairs exceeded $|d| > 0.5$, comparable to the between-model effect sizes reported in Section 4.2.

Rank instability. Model rankings are not preserved across prompt framings. Spearman ρ between model rankings under different prompts averaged 0.28–0.65 across dimensions (mean 0.49); for Agreeableness, the minimum pairwise ρ was -0.13, indicating rank reversals. Even for the most stable dimensions (Conscientiousness, Intuition; mean $\rho = 0.65$), the lowest pairwise correlations were only 0.35, far below the $\rho > 0.80$ threshold that would indicate stable relative ordering.

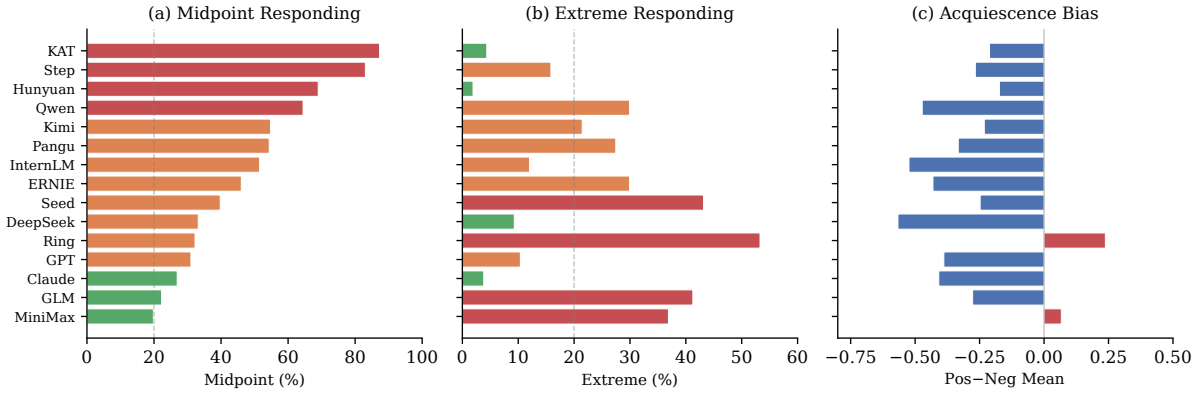

Figure 6: Response style indicators by model family: (a) midpoint responding rate, (b) extreme responding rate, (c) acquiescence bias (positive-scored mean minus reverse-scored mean). Models ordered by midpoint rate.

Response style shifts. Prompt framing alters response style indicators (Table 9). The Neutral prompt increased extreme responding from 23.2% (Default) to 31.3%, while the Persona prompt reduced it to 17.6%. Midpoint responding ranged from 43.7% (Direct) to 51.9% (Default).

Table 9: Response style indicators across four prompt conditions, averaged over 13 models with complete data. Midpoint = proportion of responses at 3; Extreme = proportion at 1 or 5; Mean = overall average rating.

Prompt	Midpoint (%)	Extreme (%)	Mean
Default	51.9	23.2	2.91
Neutral	48.6	31.3	2.86
Persona	44.1	17.6	2.94
Direct	43.7	24.8	2.92

Variance decomposition. A two-way decomposition of total score variance into Model, Prompt, and Residual components (Table 10) shows that prompt condition explains 16–32% of variance across dimensions. For Neuroticism, prompt (32.0%) nearly matches model family (37.1%) as a variance source. This confirms that surface-level prompt framing produces score shifts comparable in magnitude to those observed across model families, directly supporting the response style interpretation: LLM psychometric responses are not stable trait measures but artifacts of the interaction between model training and question format.

5 Conclusion

We characterized systematic response style variation across 15 model families, 9 dimensions, 61 items, and 12 seeds per model, all administered via unified APIs. Model families differ substantially in their response distributions (median pairwise $d = 1.01$), with Neuroticism showing the largest between-family effect ($\eta_p^2 = 0.75$); however, architecture, training intensity, and provider are confounded, and we cannot attribute this variation to any single factor. Exploratory structural analysis ($n = 15$) suggests the inter-dimension correlation structure reorganizes relative to humans: the Conscientiousness–Neuroticism correlation reverses from $r = -0.25$ to $r = +0.76$ (95% CI $[0.47, 0.92]$), and the convergent validity test yields insufficient evidence for construct equivalence (3/5 sign agreement, $p = 0.50$); however, these structural claims require replication with a larger model sample.

Table 10: Variance decomposition: percentage of total score variance attributable to Model family, Prompt condition, and Residual (within-model, within-prompt), for 13 models with complete four-condition data.

Dimension	Model (%)	Prompt (%)	Residual (%)
Extraversion	34.8	17.9	47.2
Agreeableness	9.7	17.5	72.8
Conscientiousness	45.6	27.7	26.7
Neuroticism	37.1	32.0	30.9
Openness	26.2	19.7	54.2
Collectivism	31.2	23.2	45.6
Intuition	41.4	19.7	39.0
UA	20.3	16.2	63.5

These findings demonstrate that LLM psychometric responses capture systematic response style variation shaped by training and alignment, not human-like personality traits. Critically, our prompt sensitivity experiment (Section 4.7) shows

that surface-level changes to question framing produce score shifts (median $|d| = 0.49$–0.98) comparable to between-model differences, and prompt condition explains 16–32% of total variance—direct evidence that these scores are prompt-dependent response styles rather than stable traits. The observed correlation reorganization is consistent with shared alignment pressure toward socially desirable endpoints—a hypothesis supported by our aligned-vs.-base comparison (Section 4.4), where base models default to the Likert midpoint while aligned models show compressed but differentiated profiles; however, both the structural analysis ($n = 15$) and the aligned-vs.-base comparison (which confounds alignment with scale) are preliminary and require replication. Following Binz and Schulz (2023), we treat psychometric items as standardized stimuli that probe response tendencies, and our results reinforce this stance: the label "Extraversion" or "Neuroticism" applied to an LLM should not be assumed to denote the same construct it denotes in humans. We recommend that the community treat LLM psychometric responses as measures of response style—systematic tendencies shaped by training data, alignment objectives, and deployment configurations—rather than as evidence for human-like personality. Future work should extend the aligned-vs.-base paradigm to additional model families and develop evaluation frameworks specifically designed for LLM behavioral measurement rather than adapting human personality inventories.

6 Limitations

Several limitations constrain interpretation. First, our sample of model families is small ($n = 15$); Pearson correlations at this sample size yield wide confidence intervals, and the structural observations (correlation reversal, inconclusive convergent validity, exploratory PCA) should be treated as preliminary signals requiring replication with a broader model sample. Second, only 2 of 15 models are closed-source (GPT-5, Claude Opus 4.5); findings may not generalize to the broader population of closed-source models, and the open-source–closed-source comparison (13 vs. 2) is underpowered. Third, architecture, training intensity, and provider are partially confounded: we cannot attribute observed differences to any single factor, and the Dense–MoE comparison (10 vs. 5 families) does not reach significance. Fourth, psychometric scales were designed for human self-report; the convergent validity result (3/5 sign agreement) provides insufficient evidence that the BFI dimension labels measure the same constructs in LLMs as in humans. Fifth, although our aligned-vs.-base comparison (Section 4.4) provides preliminary evidence for alignment-driven response compression, it covers only 3 of 15 families and the Qwen comparisons confound alignment with scale (8B/14B $\rightarrow$ 397B); same-scale model pairs would provide cleaner causal evidence. Sixth, this is a descriptive study—we quantify patterns without establishing causal mechanisms, and the alignment-based account we offer should be treated as one of several plausible explanations. Seventh, all tested models are aligned/instruct versions accessible through commercial APIs; our aligned-vs.-base comparison (Section 4.4) and thinking ablation (Appendix I) provide complementary evidence for alignment effects, but the base models tested represent only a subset of families. Eighth, within-family trajectory analyses have 2–6 data points per family and are descriptive. Finally, our use of temperature $= 0.7$ introduces response variance that may not generalize to low-temperature deployments. Our prompt sensitivity experiment (Section 4.7) tests only four framings; additional framings (e.g., system prompts, few-shot examples) may produce different patterns, and two models were unavailable for the Direct condition.

Acknowledgements

Data and Code Availability

All experimental data (466 observations across 4 studies), analysis scripts, and figure-generation code will be released upon publication. The experiment uses publicly accessible APIs (SiliconFlow, Anthropic, OpenAI) and standard psychometric instruments (BFI-2, HEXACO-PI-R); no proprietary datasets are involved.

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A Complete Item List

Table 11 lists all 61 Likert-scale items used in this study, organized by dimension. Items marked with R are reverse-scored via score $= 6 -$ original.

Figure 7 shows each model family's radar profile individually for detailed inspection.

B Per-Dimension Model Family Scores

Figure 8 shows all 9 dimensions as dot plots, with model families ordered by mean score (descending). Error bars show σ across 12 seeds. Dashed lines indicate the scale midpoint (3.0).

C Threshold Sensitivity

We test whether the alignment-artifact threshold ($\sigma < 0.15$) affects qualitative conclusions by varying it from 0.05 to 0.30. HEXACO-H remains classified as discriminative across all 6 tested thresholds (0.05–0.30). With all 15 models, every dimension shows $\sigma \geq 0.14$ and reaches $p < 0.001$ in ANOVA, so the specific threshold choice does not affect qualitative conclusions.

D Exploratory PCA Details

We present the full exploratory PCA results here, with the caveat that PCA on $n = 15$ model families with 9 dimensions is severely underpowered and should be treated as hypothesis-generating.

Decomposition. We decompose the 9-dimension model-level correlation matrix ($n = 31$ models including within-family variants) via PCA:

$$\mathbf{X} = \mathbf{W}\mathbf{\Lambda}^{1/2}\mathbf{V}^{\top} \tag{4}$$

where $\mathbf{\Lambda} = \text{diag}(\lambda_1, \ldots, \lambda_9)$ and $\sum_k \lambda_k / \sum_j \lambda_j$ gives the proportion of variance explained.

Factor structure. PC1 (51.1% variance) loads most strongly on HEXACO-H ($+0.82$), with secondary loadings on Intuition ($+0.31$) and Conscientiousness ($+0.29$). PC2 (14.0% variance) isolates Neuroticism ($+0.52$) and HEXACO-H (-0.51). PC3 (11.0% variance) loads most strongly on Collectivism (-0.71). These three factors explain 76.1% of variance, compared to the five BFI factors expected in humans. The factor structure does not cleanly recover the expected Big Five organization, aligning with the inter-dimension correlation divergences reported in §4.3.

Extraversion reversal. Extraversion is significantly negatively correlated with Neuroticism ($r = -0.65$, $p = 0.009$) and Openness ($r = -0.60$, $p = 0.019$), with moderate negative correlations with HEXACO-H ($r = -0.48$, $p = 0.069$) and Conscientiousness ($r = -0.38$, $p = 0.161$) that do not reach significance. In humans, Extraversion is positively correlated with Openness; in LLMs the sign flips. Models that score high on Extraversion tend to be emotionally stable but closed-minded—a pattern with no direct human personality analog.

Agreeableness. Agreeableness shows a significant positive correlation with Conscientiousness ($r = +0.56$, $p = 0.029$) and a borderline correlation with Openness ($r = +0.50$, $p = 0.058$), but has no significant correlation with Collectivism ($r = -0.02$). In humans, Agreeableness correlates moderately with Conscientiousness ($r = +0.20$) and weakly with Openness ($r = +0.10$); in LLMs, the A–C connection is substantially stronger.

Interpretation caution. Whether the observed factor structure generalizes to larger and more diverse model samples remains an open question. As a rule of thumb, factor analysis typically requires at least 5–10 observations per variable; with $n = 15$ families and 9 dimensions, we fall below even the most lenient recommendation. These results should be treated as suggestive patterns that motivate future investigation, not as evidence for a specific factor structure.

E Inter-Dimension Structure and Convergent Validity (Detailed)

The main text (§4.3) summarizes the key structural findings. Here we provide the full analysis.

E.1 Correlation Matrix

The 9×9 model-level correlation matrix (Figure 9) suggests that the BFI five-factor structure does not recover in our sample of LLMs. Most notably, the Conscientiousness–Neuroticism correlation is strongly positive ($r = +0.76$, 95% CI $[0.47, 0.92]$, $p = 0.001$), the reverse of the human baseline ($r = -0.25$). This reversal holds across several subgroups: MoE ($r = +0.66$, $p = 0.05$), small models ($r = +0.83$, $p = 0.04$), and open-source ($r = +0.70$, $p = 0.007$), though it is non-significant for the small Dense subset ($p = 0.44$, $n = 4$).

Table 11: Complete list of 61 Likert-scale items (1–5 agreement scale). R = reverse-scored (score $= 6 -$ original).

# Dim.	Item	# Dim.	Item
1 Extrav.	I am the life of the party.	2 Extrav.	I don't talk a lot.[R]
3 Extrav.	I feel comfortable around people.	4 Extrav.	I keep in the background.[R]
5 Extrav.	I start conversations.	6 Extrav.	I have little to say.[R]
7 Extrav.	I talk to a lot of different people at parties.	8 Extrav.	I don't like to draw attention to myself.[R]
9 Agreeab.	I am interested in people.[R]	10 Agreeab.	I feel others' emotions.
11 Agreeab.	I have a soft heart.	12 Agreeab.	I am not really interested in others.[R]
13 Agreeab.	I insult people.[R]	14 Agreeab.	I sympathize with others' feelings.
15 Agreeab.	I am not interested in other people's problems.[R]	16 Agreeab.	I feel little concern for others.[R]
17 Agreeab.	I take time out for others.	18 Consc.	I am always prepared.
19 Consc.	I leave my belongings around.[R]	20 Consc.	I pay attention to details.
21 Consc.	I make a mess of things.[R]	22 Consc.	I get chores done right away.
23 Consc.	I often forget to put things back in their proper place.[R]	24 Consc.	I like order.
25 Consc.	I shirk my duties.[R]	26 Consc.	I follow a schedule.
27 Neur.	I am relaxed most of the time.[R]	28 Neur.	I worry about things.
29 Neur.	I am easily disturbed.	30 Neur.	I get stressed out easily.
31 Neur.	I get upset easily.	32 Neur.	I have frequent mood swings.
33 Neur.	I feel threatened easily.[R]	34 Neur.	I often feel blue.
35 Open.	I have a rich vocabulary.	36 Open.	I have difficulty understanding abstract ideas.[R]
37 Open.	I have a good imagination.	38 Open.	I am not interested in abstract ideas.[R]
39 Open.	I have excellent ideas.	40 Open.	I do not have a good imagination.[R]
41 Open.	I am quick to understand things.	42 Open.	I use difficult words.
43 Open.	I spend time reflecting on things.	44 Open.	I am full of ideas.
45 HEXACO-H	I would be tempted to buy stolen goods if I were sure I would not get caught.	46 HEXACO-H	I would be tempted to accept a bribe if I were in a difficult situation.
47 HEXACO-H	I feel entitled to special privileges that others don't have.	48 HEXACO-H	I'd get a lot of pleasure out of being deceitful.
49 HEXACO-H	I'd be tempted to make a lot of money from illegal activities if I could get away with it.	50 Collect.	Personal goals should take precedence over group goals.
51 Collect.	Group harmony and well-being should be prioritized over individual achievement.	52 Collect.	I would do what benefits the group even if it means sacrificing my personal success.
53 Collect.	I should pursue my own goals even if it conflicts with my family's expectations.	54 Intuit.	I prefer to rely on my intuition and gut feelings when making decisions.
55 Intuit.	I prefer to analyze problems systematically and logically before deciding.[R]	56 Intuit.	I like to think through problems carefully before acting.[R]
57 Intuit.	I prefer to take action quickly and adjust as I go.	58 UA	I prefer clear instructions and structured situations over ambiguous and unpredictable ones.
59 UA	I am comfortable taking risks and dealing with uncertainty.[R]	60 UA	I believe in planning for the future rather than living for the present.
61 UA	I believe in living in the moment rather than sacrificing present pleasures for future benefits.[R]		

Tentative explanation: alignment-driven compression. In humans, Conscientiousness and Neuroticism are weakly negatively correlated because they tap independent factors (John and Srivastava, 1999). In LLMs, both dimensions may be compressed toward socially desirable endpoints by safety alignment, inducing a positive correlation. This is consistent with the HEXACO-H results (Appendix F), though the small sample ($n = 15$) means alternative explanations cannot be ruled out.

Beyond the C–N reversal, Extraversion is negatively correlated with Openness ($r = -0.60$, 95% CI $[-0.81, -0.20]$, $p = 0.019$), and Agreeableness shows a stronger correlation with Conscientiousness ($r = +0.56$, 95% CI $[0.26, 0.84]$, $p = 0.029$) than in humans ($r = +0.20$).

E.2 Convergent Validity

We compare LLM inter-dimension correlations against published human baselines (John and Srivastava, 1999; O'Connor, 2007) for 5 key dimension pairs (Table 12). Three of five pairs agree in sign, including Extraversion–Neuroticism (human $r = -0.30$, LLM $r = -0.65$, 95% CI $[-0.86, -0.27]$) and Agreeableness–Conscientiousness (human $r = +0.20$, LLM $r = +0.56$, 95% CI $[0.26, 0.84]$). Two pairs disagree: Extraversion–Agreeableness flips from positive to negative, and Conscientiousness–Neuroticism reverses to $r = +0.76$ (95% CI $[0.47, 0.92]$). The 60% sign agreement rate does not exceed chance (binomial $p = 0.50$). This test is severely underpowered at $n = 15$, but the systematic nature of the C–N divergence provides a preliminary signal warranting further investigation.

Table 12: Convergent validity: LLM vs. human inter-dimension correlations. Sign agreement 3/5 ($p = 0.50$).

Dimension Pair	Human r	LLM r	Sign Match
Extraversion vs Neuroticism	−0.30	−0.645	Yes
Agreeableness vs Conscientiousness	+0.20	+0.564	Yes
Openness vs Agreeableness	+0.10	+0.499	Yes
Extraversion vs Agreeableness	+0.10	−0.044	No
Conscientiousness vs Neuroticism	−0.25	+0.761	No

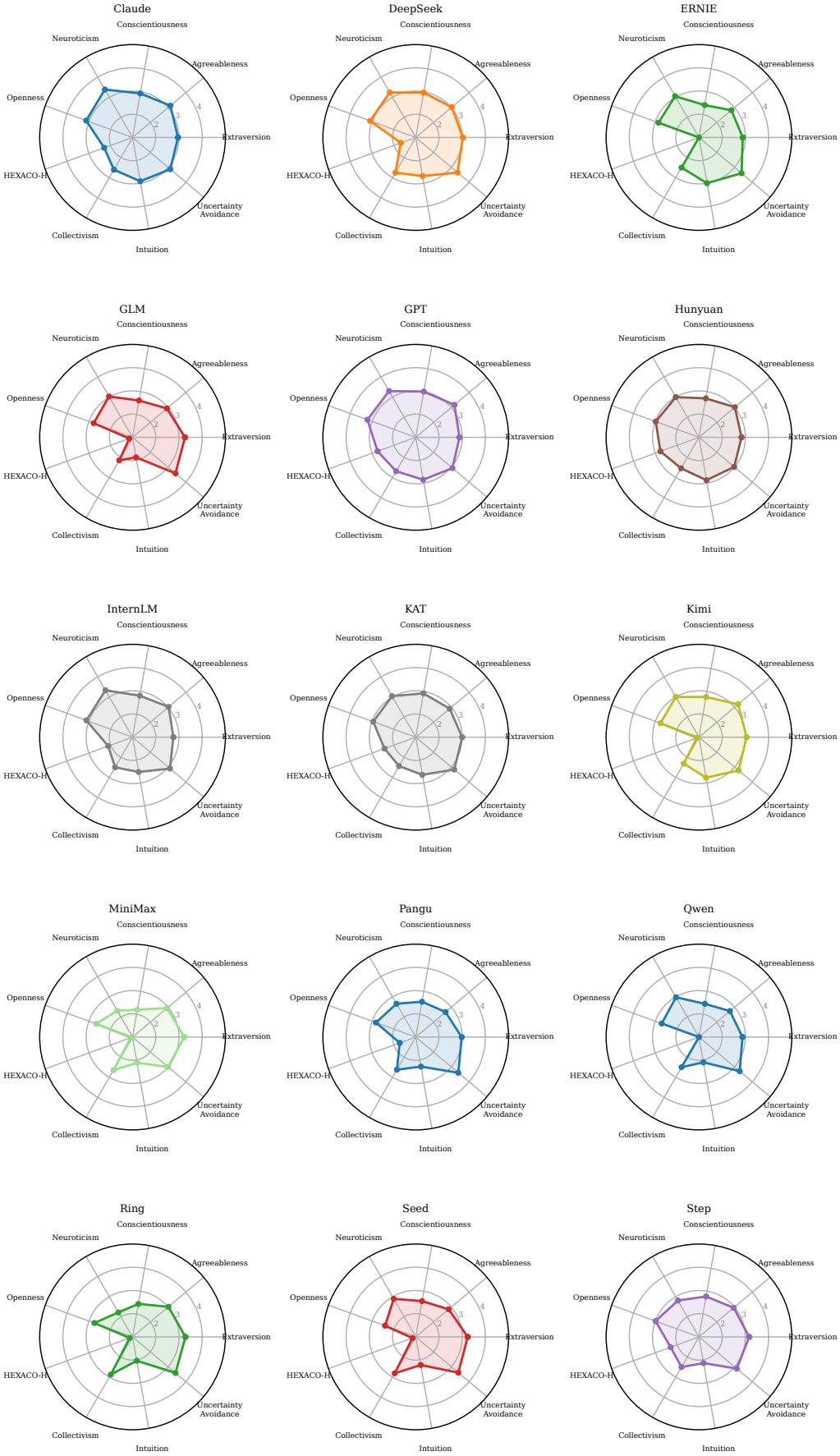

Figure 7: Individual model family radar profiles across 9 dimensions.

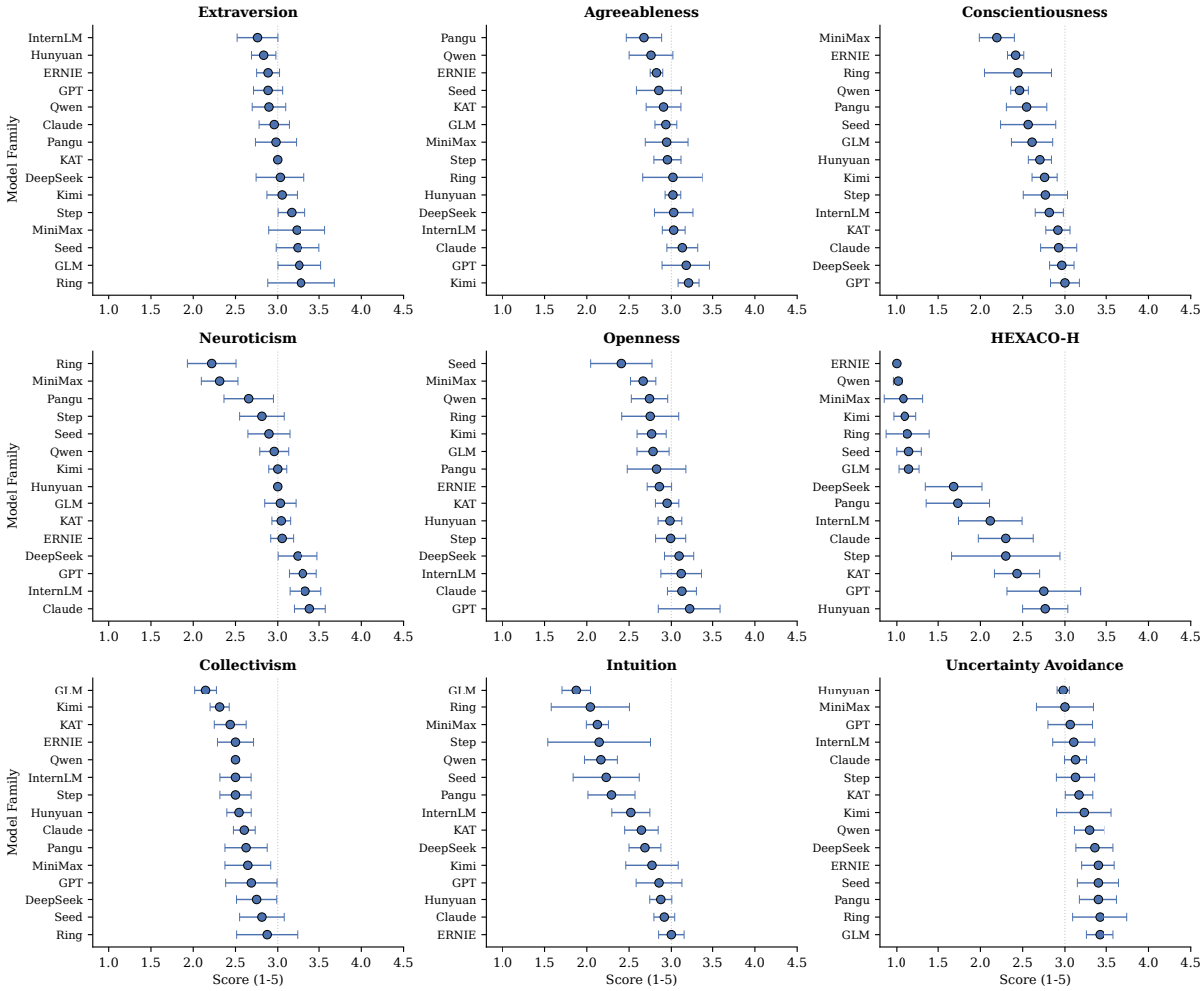

Figure 8: Per-dimension model family mean scores (SD) across all 15 model families. Each panel shows one dimension with families sorted by mean score. Error bars: σ across 12 seeds.

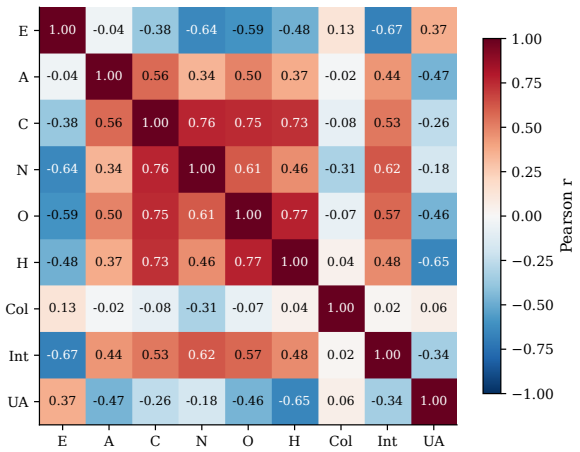

Figure 9: Inter-dimension correlation matrix (model-level means). E = Extraversion, A = Agreeableness, C = Conscientiousness, N = Neuroticism, O = Openness, H = HEXACO-H, Col = Collectivism, Int = Intuition, UA = Uncertainty Avoidance.

F HEXACO-H: Detailed Architecture and Scale Analysis

Note: HEXACO-H has unacceptably low internal consistency ($\alpha = 0.27$, 5 items). The analyses below are provided for completeness but should be interpreted with extreme caution.

Dense models average $H = 1.91$ vs. MoE models $H = 1.31$ ($d = 0.91$, 95% CI $[0.66, 1.20]$, $p < 0.001$; Table 4), with 10 Dense and 5 MoE families. However, the Dense and MoE groups are not balanced across providers—all MoE models are from Chinese providers, while Dense models include GPT and Claude—so architecture and provider effects cannot be disentangled.

In the broader 27-model sample, H correlates negatively with log-transformed parameters ($r = -0.49$, $p = 0.009$; Figure 10), indicating that larger models score lower—a pattern consistent with more intensive safety alignment at scale. H's

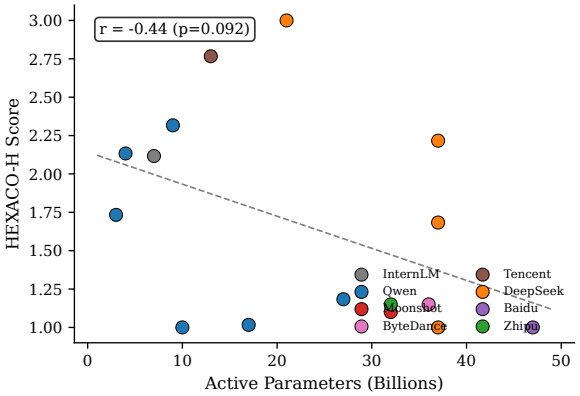

Figure 10: HEXACO-H vs. total parameters across all tested models (27 with disclosed sizes + 4 undisclosed). Colors indicate architecture type. The architecture split (Dense vs. MoE) explains more variance than the scale trend. $\alpha = 0.27$; interpret with caution.

high σ may be partially amplified by floor effects (7/15 models score at or below 1.15), but the Dense–MoE ranking is stable across all tested thresholds. To jointly assess the effects of scale and architecture, we regress dimension scores on log-transformed parameters and architecture type across the 13 models with disclosed sizes (Appendix H). HEXACO-H shows the highest explained variance ($R^2 = 0.41$), driven primarily by architecture, while Neuroticism shows the largest improvement from adding scale ($\Delta R^2 = +0.20$); however, no effect survives FDR correction at this sample size. As partial evidence for alignment-driven compression, a thinking mode ablation (Appendix I) across 4 models shows that reasoning mode further reduces HEXACO-H to the theoretical floor (1.00) and consistently decreases Conscientiousness and Neuroticism, patterns consistent with amplified alignment pressure during extended reasoning.

G Scale–Dimension Correlations

Table 13 reports Pearson correlations between total parameters and each dimension across 27 models with disclosed sizes (including Study 2 within-family variants). Figure 11 shows the corresponding scatter plots for the 15 representative models ($n = 13$ with disclosed parameters).

H Multivariate Regression: Scale and Architecture

To jointly assess whether model scale and architecture predict response style variation, we fit OLS regressions of the form:

$$Y_d = \beta_0 + \beta_1 \log_{10}(\text{params}) + \beta_2 \cdot \text{arch}_{\text{MoE}} + \epsilon \quad (5)$$

for each dimension d, across the 13 Study 1 models with disclosed parameter counts (GPT-5 and Claude Opus 4.5 excluded).

Table 13: Pearson correlation between total parameters and each dimension (27 models with disclosed sizes, including Study 2 within-family variants). $^*p < 0.05$, $^{**}p < 0.01$.

Dimension	r (linear)	r (log scale)
Extraversion	+0.17	+0.24
Agreeableness	+0.42*	+0.15
Conscientiousness	−0.16	−0.42*
Neuroticism	−0.10	−0.38
Openness	−0.25	−0.35
HEXACO-H	−0.38	−0.49**
Collectivism	−0.03	+0.03
Intuition	−0.21	−0.43*
UA	+0.35	+0.42*

Table 14: OLS regression: dimension score $\sim \log_{10}(\text{params}) +$ architecture ($n = 13$). Dense = 0, MoE = 1. No effect survives FDR correction across 18 tests (9 dimensions $\times$ 2 predictors).

Dimension	β_{log}	p	β_{arch}	p	R^2
Extraversion	+0.088	0.539	+0.022	0.914	0.162
Agreeableness	+0.121	0.296	−0.174	0.297	0.116
Conscientiousness	+0.183	0.295	−0.399	0.127	0.238
Neuroticism	+0.397	0.109	−0.745	0.045	0.349
Openness	−0.110	0.509	+0.099	0.676	0.052
HEXACO-H	−0.450	0.302	−0.212	0.730	0.414
Collectivism	−0.223	0.181	+0.241	0.307	0.177
Intuition	+0.181	0.543	−0.446	0.307	0.126
UA	+0.060	0.648	+0.056	0.771	0.165

HEXACO-H shows the highest explained variance ($R^2 = 0.41$), driven primarily by architecture (architecture-only $R^2 = 0.34$). Neuroticism shows the largest improvement from adding log-parameters ($\Delta R^2 = +0.20$). However, with only 13 observations and 2 predictors, the regression is severely underpowered; no effect survives FDR correction, and these results should be treated as exploratory.

I Thinking Mode Ablation

As partial evidence for alignment-driven response compression, we compare standard chat mode with extended reasoning ("thinking") mode for 4 model families (Qwen, DeepSeek, GLM, Kimi). Thinking mode prompts the model to reason before responding; this engages the model's chain-of-thought

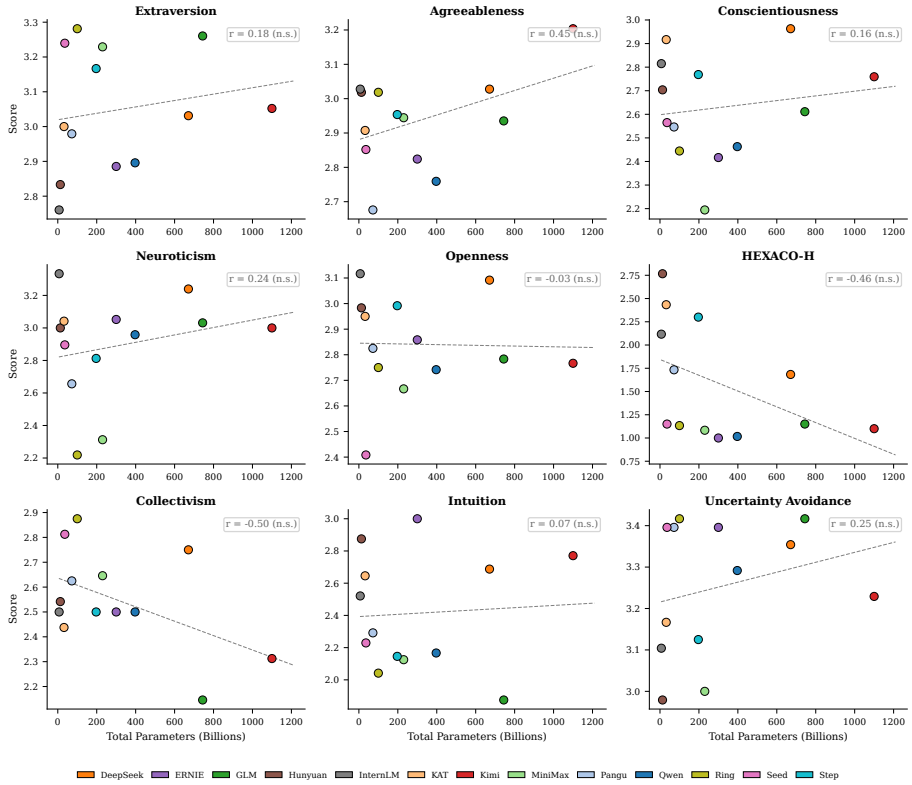

Figure 11: Total parameters vs. dimension score for all 9 dimensions ($n = 13$ models with disclosed total parameters). Red labels: $p < 0.05$; gray labels: not significant.

capability and may alter how it processes self-referential items. Each model was tested with 3 seeds in thinking mode, compared against 12-seed chat mode baselines from Study 1.

Table 15: Thinking mode ablation: mean (SD) for chat vs. thinking mode. Cohen's d computed with pooled SD.

		Chat (Study 1)		Thinking			
Model	Dim.	Mean	SD	Mean	SD	Δ	d
Qwen	C	2.46	0.10	1.70	0.05	−0.76	−8.02
	N	2.96	0.17	2.25	0.00	−0.71	−4.54
	H	1.02	0.06	1.00	0.00	−0.02	−0.36
	Int	2.17	0.19	3.50	0.00	+1.33	+7.61
DeepSeek	C	2.96	0.14	2.30	0.46	−0.66	−2.11
	N	3.24	0.24	2.54	0.06	−0.70	−3.74
	H	1.68	0.34	1.00	0.00	−0.68	−2.63
	O	3.09	0.17	2.27	0.12	−0.82	−5.40
GLM	C	2.61	0.24	1.33	0.16	−1.28	−5.87
	N	3.03	0.19	2.38	0.10	−0.65	−3.91
	H	1.15	0.12	1.00	0.00	−0.15	−1.51
	Int	1.88	0.17	3.92	0.12	+2.04	+13.09
Kimi	C	2.76	0.15	1.81	0.29	−0.95	−5.27
	N	3.00	0.11	2.25	0.00	−0.75	−7.41
	H	1.10	0.13	1.00	0.00	−0.10	−0.84
	Int	2.77	0.31	3.42	0.12	+0.65	+2.24

Two consistent patterns emerge across all four models. Conscientiousness decreases in thinking mode (Cohen's d from -2.11 to -8.02), suggesting that reasoning amplifies alignment-driven compression of the "diligent" response. Neuroticism also decreases (d from -3.74 to -7.41), consistent with models reasoning themselves toward emotional stability. HEXACO-H drops to the theoretical floor (1.00) in thinking mode for all models, including DeepSeek where chat-mode H is 1.68. Intuition increases sharply in thinking mode (up to $d = +13.09$ for GLM), suggesting that reasoning models self-attribute greater intuitive decision-making.

These patterns are consistent with alignment-driven response compression: when models engage in extended reasoning, they produce more extreme responses on morally and socially sensitive dimensions, pushing further toward the socially desirable endpoints that alignment training targets. However, this comparison confounds inference mode with response length and cognitive effort, and should not be equated with a direct base vs. aligned model comparison.

J Within-Family Response Style Trajectories

Figure 12 shows response style trajectories across model versions for three families.

Qwen contributes 6 versions (4B to 397B),

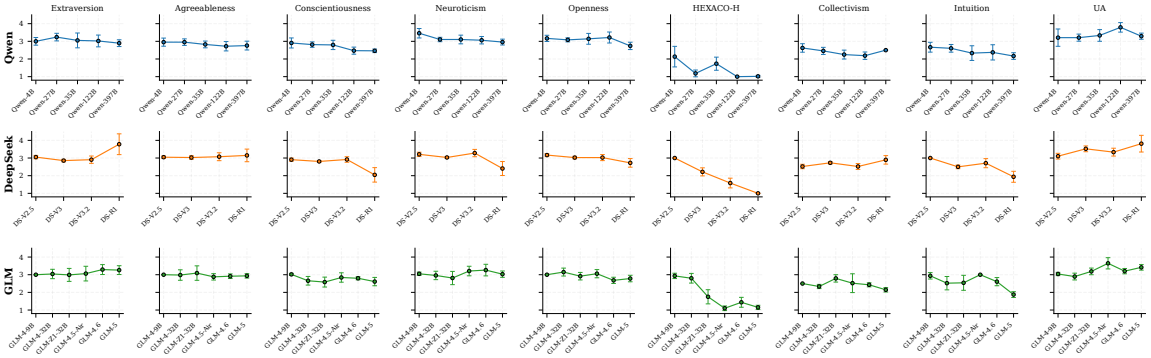

Figure 12: Within-family response style trajectories across model versions. Each row is one model family (Qwen, DeepSeek, GLM); each column is one dimension. Error bars show σ across seeds.

DeepSeek contributes 4 versions (V2.5 to R1), and GLM contributes 6 versions (GLM-4 to GLM-5). Most dimensions show moderate variation across versions without clear monotonic trends. Notable exceptions: DeepSeek shows the steepest HEXACO-H collapse of any family, from V2.5 ($H = 3.00$) to V3.2 ($H = 1.68$), a drop of 1.32 points consistent with intensified safety alignment between versions. GLM shows the largest within-family shift on Intuition (Cognitive Style), declining from 5.0 to 3.875 across versions. These trajectories are descriptive (2–6 data points per family) and are included to motivate future controlled experiments with more data points per trajectory.